

ONKAR KOTULE

Software Development Engineer (SDE-2) | Backend & Full-Stack | MERN | Azure | Mendix

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PROFESSIONAL SUMMARY

Software Development Engineer with 4+ years of experience building scalable backend frontend systems and full-stack applications across enterprise and startup environments. Strong expertise in API design, data modeling, and performance optimization, with hands-on experience designing transaction-safe and real-time systems. Proficient in developing high-performance, production-grade architectures using Node.js, MongoDB, React, Next.js, and cloud platforms, TypeScript, with a focus on scalability, consistency, and clean system design. Proven ability to own features end-to-end and build reliable, scalable systems in production. production, with experience in frontend development, real-time streaming, and UI design. Experienced in agile development methodologies and familiar with scrum principles.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Java, Python

Frontend: React.js, Next.js, React Native, Expo, Tailwind CSS, Redux Toolkit, TanStack Query Query, Framer Motion, Shadcn/UI, Zustand, Websocket

Backend: Node.js, Express.js, MongoDB, PostgreSQL, REST APIs, GraphQL, JWT, OAuth

Cloud & DevOps: Azure DevOps, CI/CD, Docker, Git, GitHub, Firebase, Appwrite

Testing: Jest, Basic Unit Testing, React Testing Library

AI/GenAI: langGraph, LLM Integration, Prompt Engineering, Agentic Workflows

Agile & Scrum: Agile, Scrum

PROFESSIONAL EXPERIENCE

Associate Consultant (Software Engineer – Backend & Full-Stack)

Jul 2023 - Present

Capgemini

- Designed and developed enterprise-scale applications across backend and frontend, optimizing state management, data flow, and UI performance for complex workflows.
- Architected backend integrations and REST API layers for scalable communication across services services, using GraphQL and TypeScript.
- Built systems with role-based access control (RBAC), auditability, and secure data flows. flows, using JWT and OAuth.

- Owned features end-to-end, from requirement design and development to production deployment and post-release ownership.

AI-Powered Investment Insights System

- Designed LLM-based pipelines to generate structured investment reports
- Implemented prompt engineering and response validation for consistent and reliable outputs
- Built dynamic frontend components to render UI based on API-driven responses, using React and Next.js
- Integrated AI-generated insights into applications for real-time user consumption, with a focus on UI design and real-time streaming.

Export Control system

- Converted rigid logic into config-driven architecture
- Improved maintainability and reduced manual effort.

ShareSpace

- Hierarchical file storage with optimized data modeling
- Secure upload/download pipelines with access control
- Soft delete (trash), restore & lifecycle management

Dangerous Goods Transport System

- Slot booking system with workflow orchestration
- Real-time notifications and approval pipelines

Software Developer (Backend & Full-Stack)

Jun 2021 - Oct 2022

BITCS | DigitalFlow | KnockOnce

- Built scalable backend services using Node.js, Express, and MongoDB, designing modular REST APIs with validation and error handling.
- Developed frontend systems using Next.js and Redux Toolkit, implementing optimized state management and efficient data fetching.
- Optimized frontend performance using: - Efficient state management and component reusability (React, Redux Toolkit). - Data fetching and caching strategies (TanStack Query). - Reduced unnecessary re-renders and improved UI responsiveness.
- Improved system performance through MongoDB indexing, query optimization, and efficient schema design.

Expense management system

- Designed and developed a production-grade expense management system handling multi-user financial transactions

- Implemented: o Transaction-safe balance updates ensuring data consistency across users o Real-time expense tracking and settlement logic
- Designed optimized data models and aggregation pipelines for analytics and reporting
- Enhanced performance using compound indexes for high-frequency queries
- Built the system with focus on scalability, concurrency handling, and data integrity integrity.

Additional Systems Built

- Built real-time communication features using event-driven architecture (Socket.IO)
- Developed modular frontend systems with optimized data fetching & caching caching.

CERTIFICATIONS

- Microsoft Certified: Azure Developer Associate (AZ-204)
- Microsoft Certified: Azure AI Engineer Associate (AI-104)
- Microsoft GenAI Practitioner (AI-900)

EDUCATION

Bachelor of Engineering (B.E.)

Bharati Vidyapeeth College of Engineering, Pune

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